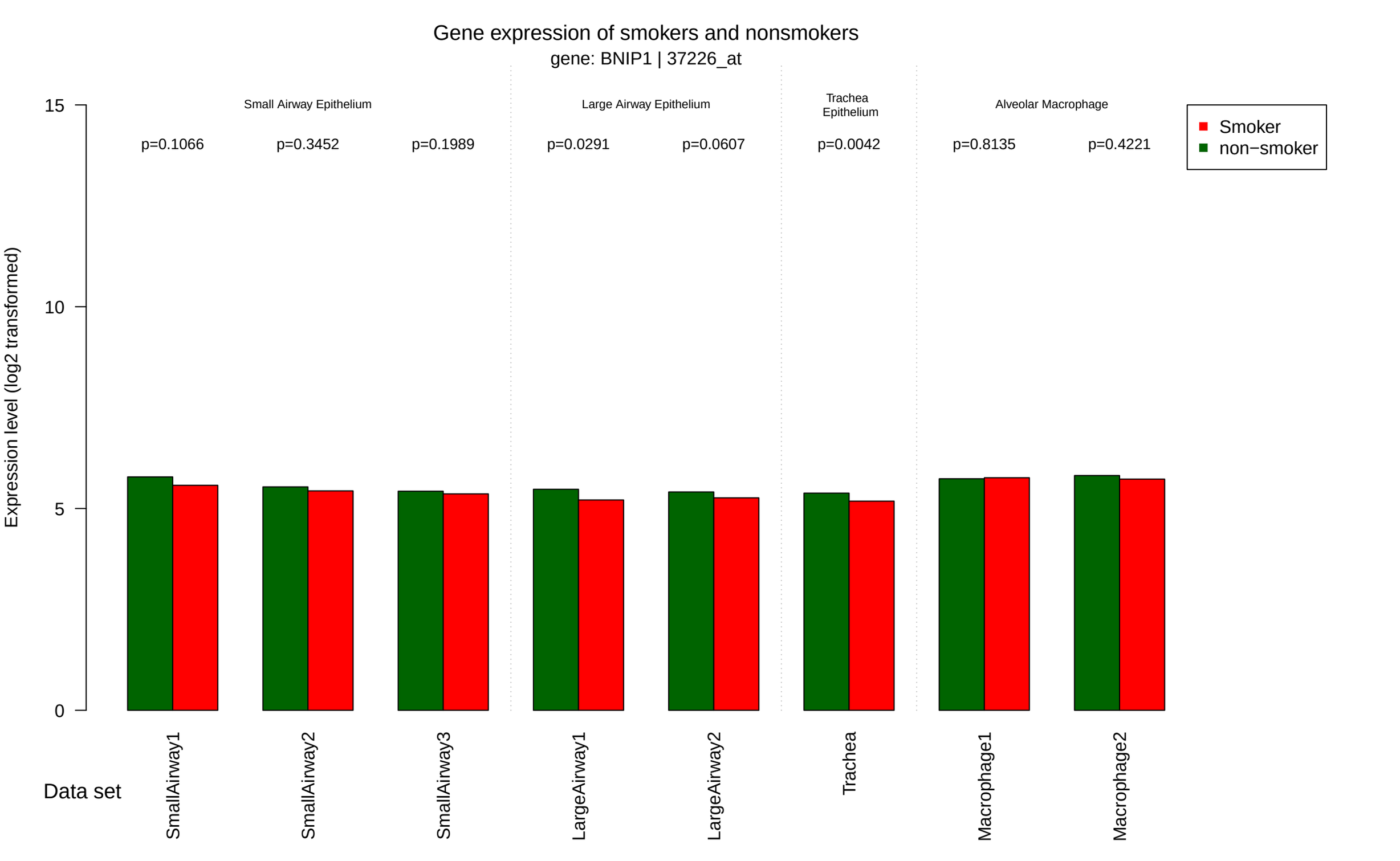
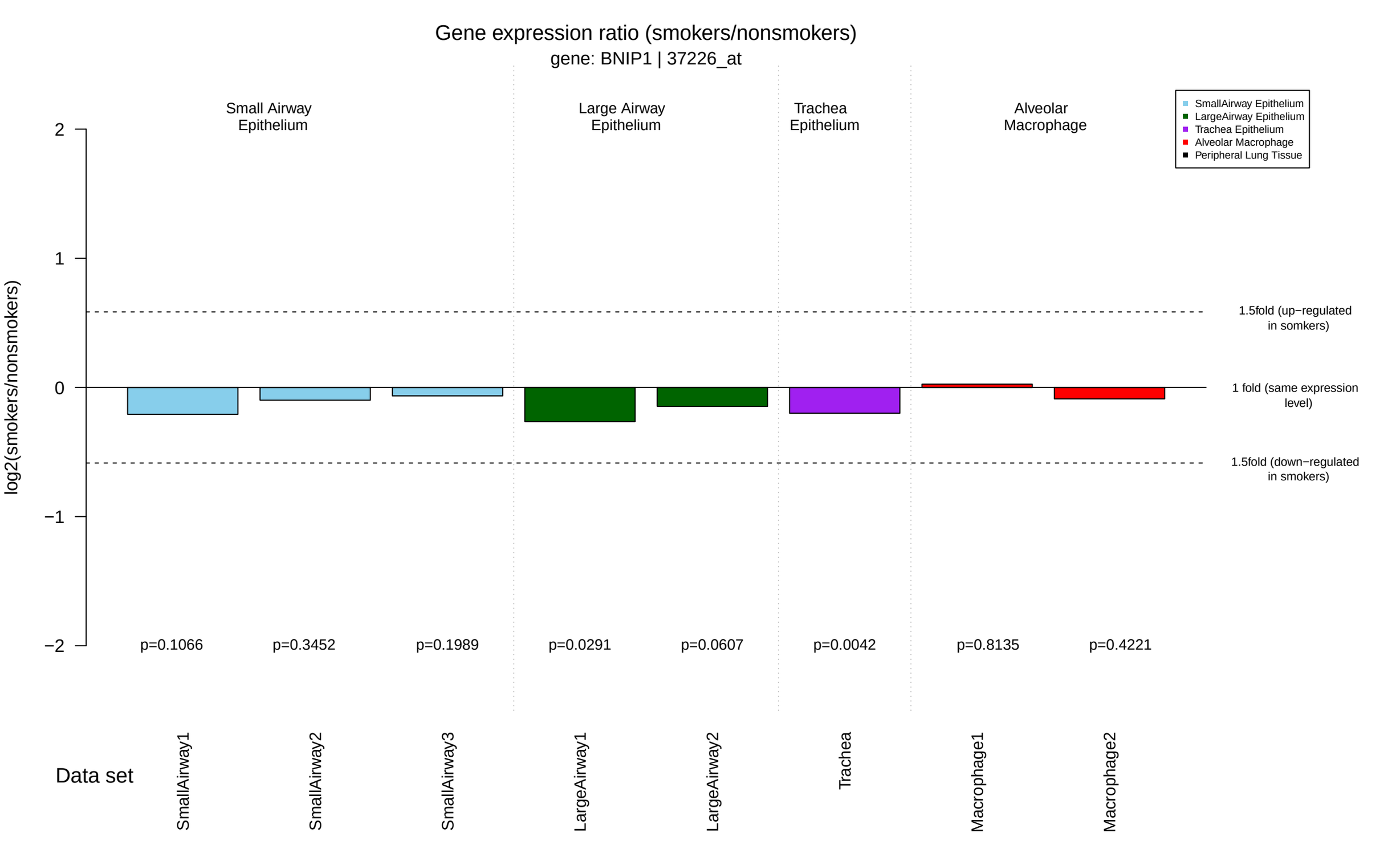
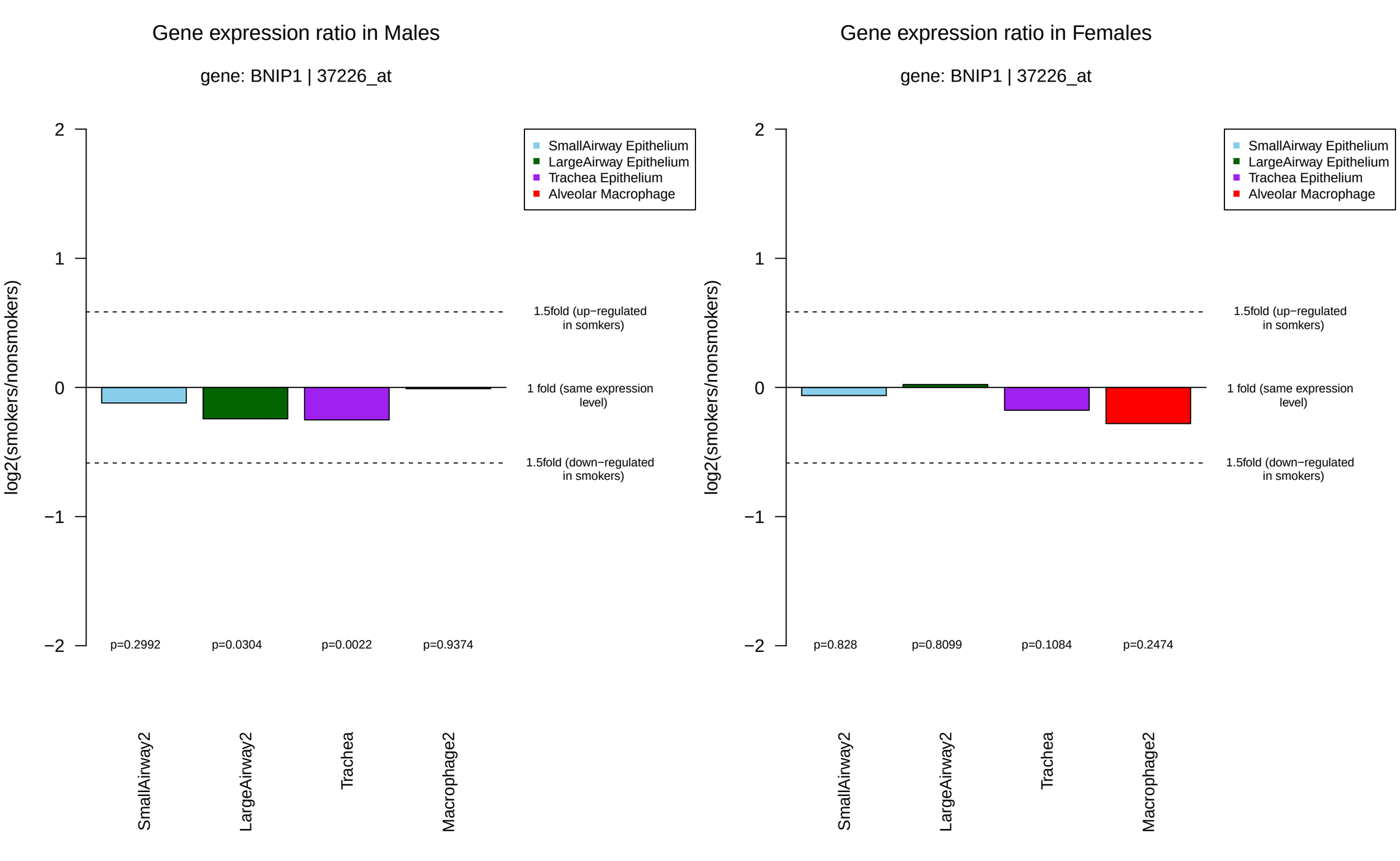
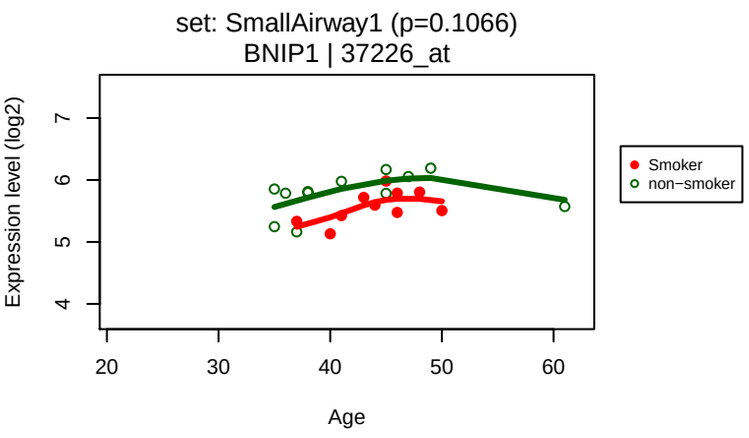
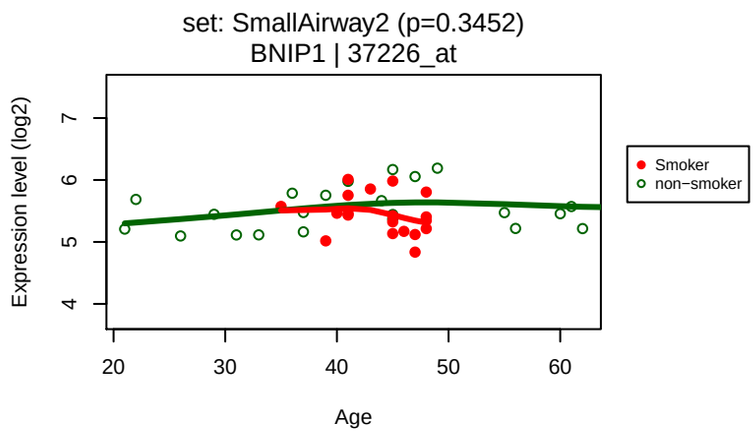
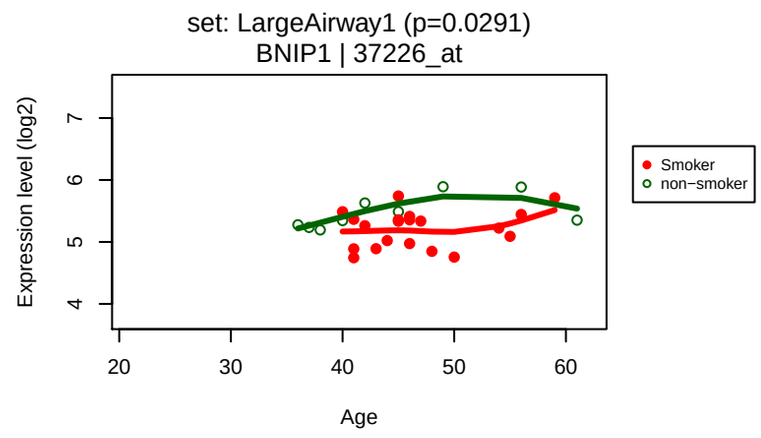

Gene expression of smokers and nonsmokers
gene: BNIP1 | 37226_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1066
p=0.3452
p=0.1989
p=0.0291
p=0.0607
p=0.0042
p=0.8135
p=0.4221
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

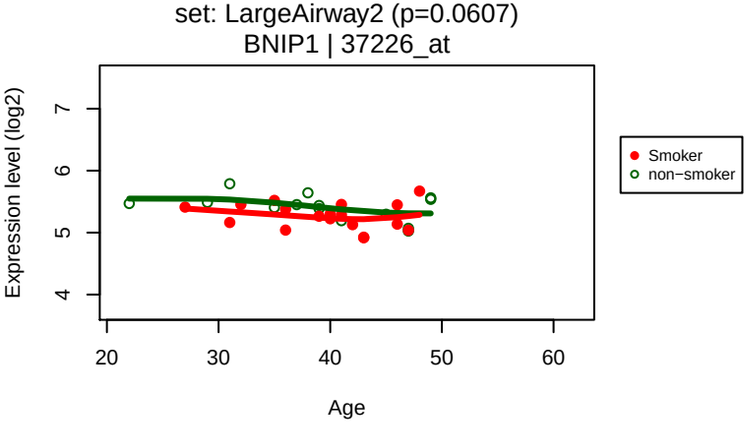

Gene expression ratio (smokers/nonsmokers)
gene: BNIP1 | 37226_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1066
p=0.3452
p=0.1989
p=0.0291
p=0.0607
p=0.0042
p=0.8135
p=0.4221
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

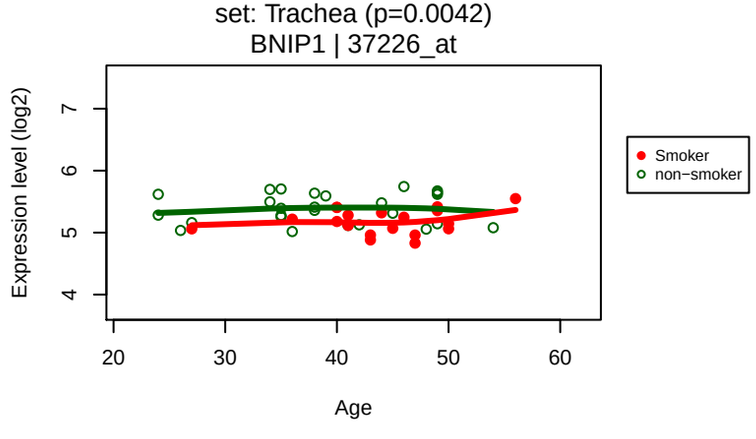

Gene expression ratio in Males
gene: BNIP1 | 37226_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2992
p=0.0304
p=0.0022
p=0.9374
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BNIP1 | 37226_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.828
p=0.8099
p=0.1084
p=0.2474
SmallAirway2
LargeAirway2
Trachea
Macrophage2

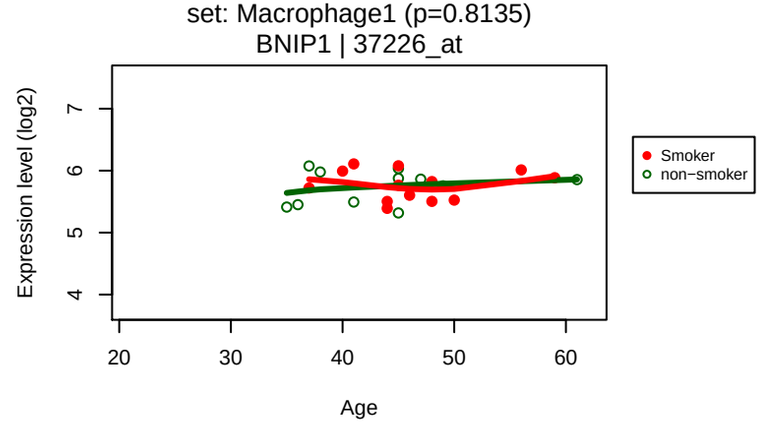

set: SmallAirway1 (p=0.1066)
BNIP1 | 37226_at
Expression level (log2)
Age
Smoker
non−smoker

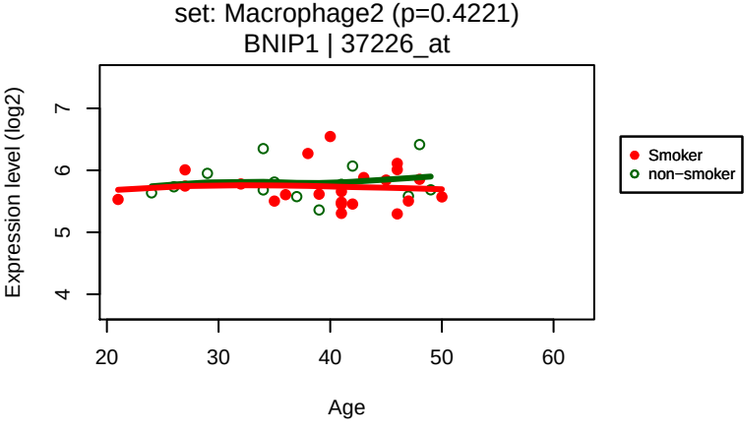

set: SmallAirway2 (p=0.3452)
BNIP1 | 37226_at
Expression level (log2)
Age
Smoker
non−smoker

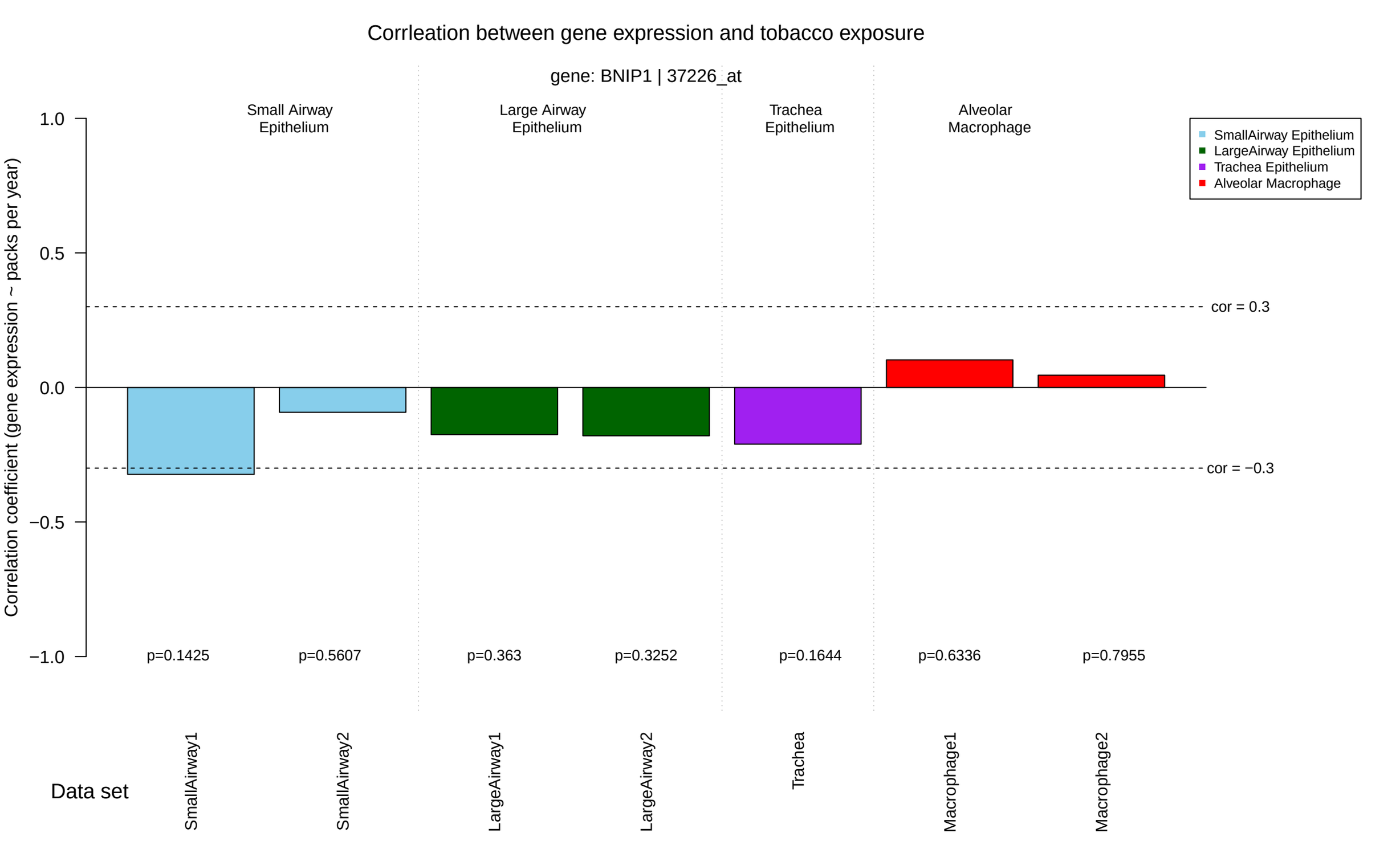

set: LargeAirway1 (p=0.0291)
BNIP1 | 37226_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0607)
BNIP1 | 37226_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0042)
BNIP1 | 37226_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8135)
BNIP1 | 37226_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4221)
BNIP1 | 37226_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BNIP1 | 37226_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1425
p=0.5607
p=0.363
p=0.3252
p=0.1644
p=0.6336
p=0.7955
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2